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DATA DESCRIPTOR

Disaster Storylines and Knowledge Graphs from Global News with Large Language Models and Retrieval-Augmented Generation

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We present a dataset of over 3,000 global disaster events from 2014 to 2024, derived from the Emergency Events Database (EM-DAT). Events are extracted from news using a pipeline combining Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) for semantic extraction. The corpus is the Europe Media Monitor (EMM), aggregating content from millions of news outlets. For each event, structured storylines are automatically generated, summarizing hazard characteristics, drivers, impacts, and responses, and transformed into knowledge graphs. This enables analysis of relationships, inter-hazard dynamics, and human-environment interactions often missed in traditional records. A small subset of knowledge graphs was evaluated by domain experts in a workshop, while a larger sample of extracted triplets was independently assessed to quantify precision and inter-annotator agreement. The dataset supports retrospective analysis and multi-hazard risk assessment, complementing resources like the Hazard Information Profiles (HIPs). All data, code, and workflows are openly available, with an interactive dashboard for exploration. This resource advances data-driven approaches to disaster scenario modeling, impact analysis, and decision support in disaster risk management.

Background & Summary

Natural hazards and extreme events pose escalating challenges for global disaster risk management (DRM), not only because of their increasing frequency and severity but also due to the dynamic and cascading effects they can produce. These include climate-related hazards such as hurricanes, floods, and wildfires, as well as geological hazards like earthquakes and landslides—each capable of severely disrupting human populations, infrastructure, and ecosystems. International frameworks such as the Sendai Framework for Disaster Risk Reduction and the United Nations Office for Disaster Risk Reduction (UNDRR) Hazard Information Profiles (HIPs)¹ have advanced understanding of individual hazards. However, significant knowledge gaps remain in systematically representing interactions between hazards and the socioeconomic vulnerabilities that amplify their impacts^{2–4}. These challenges become particularly acute when multiple hazards occur in sequence or combination, producing compound or cascading impacts that existing disaster data systems struggle to capture^{5–7}. Initiatives such as “Early Warnings for All” (EW4ALL)⁸ underscore the urgency of more integrated and data-informed approaches to hazard monitoring, risk assessment, and response planning.

Recent advances in natural language processing, particularly the emergence of large language models (LLMs)^{9,10}, open new opportunities to address these challenges by extracting insights from unstructured sources. In disaster contexts, where traditional datasets are often delayed, incomplete, or unavailable, textual sources such as news reports, field assessments, and social media posts can provide valuable complementary information^{11–15}. These texts capture qualitative and quantitative details on impacts, exposed populations, local vulnerabilities, coping strategies, and cascading effects—details that fill critical blind spots in existing disaster databases^{16–19}. When paired with retrieval-augmented generation (RAG)²⁰—a method that grounds large

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language model outputs in external knowledge sources-LLMs can transform this unstructured content into coherent, factually grounded narratives, enabling structured representation of events, entities, and relationships at scale. This approach supports deeper exploration of the human-hazard interface and complex risk interdependencies across spatial and temporal scales.

In this study, we present a dataset²¹ of over 3,000 global disaster events from 2014–2024, derived from the Europe Media Monitor (EMM), a multilingual media aggregation platform that curates articles from thousands of vetted news sources and includes more than 1.5 billion articles worldwide²². We focus on English-language articles from this corpus and process them using a RAG pipeline combining the power of LLMs for text generation and the semantic extraction of factual knowledge from the unstructured news text. For each disaster event recorded in the Emergency Events Database (EM-DAT)¹⁶, the system produces a structured storyline that summarizes key aspects including hazard types, main drivers, direct and indirect impacts, affected sectors and populations, and response actions taken. These storylines are further transformed into Knowledge Graphs (KGs)²³ that capture multi-hazard dynamics, inter-event relationships, and interactions between physical hazards and societal responses.

In recent years, the research community has successfully used KGs for creating semantically enriched representations of data in various domains²⁴. KGs are data structures that describe the key entities in a domain and their interconnections, presenting information in a format accessible and interpretable by both machines and humans²⁵. The relationship between two entities is typically formalized as a triple in the format of *<subject, predicate, object>* (e.g., *<India-Pakistan heatwave, cause, 87 deaths>* or *<Ebola outbreak, occur, Congo>*). KGs are recognized for their ability to organize data in a predefined and semantically meaningful manner, offering significant support to AI systems across multiple domains²⁶. In our application context, by representing disaster events as interconnected networks of entities and relationships, the obtained dataset provides a means to examine complex risk patterns that are typically underrepresented in classical sources^{1,16}.

To assess the reliability and usefulness of the extracted disaster KGs, a small sample was independently reviewed and discussed by domain experts during a validation workshop (see <https://civil-protection-knowledge-network.europa.eu/news/how-can-ai-strengthen-disaster-preparedness-europe>) involving civil protection authorities and disaster analysts. Although this validation covered only a limited set of events, it nonetheless provided high-quality expert feedback that confirmed the relevance, plausibility, and potential value of the extracted information for supporting situational awareness and retrospective analysis. In addition, a larger quantitative validation was conducted on 1,000 randomly sampled triplets extracted from the generated knowledge graphs, stratified to reflect the distribution of disaster types in the full dataset. These triplets were independently annotated by experts to assess factual correctness, enabling the estimation of precision and inter-annotator agreement. The complete dataset²¹, along with code, extraction templates, and processing workflows, is openly available on Zenodo at <https://doi.org/10.5281/zenodo.18598183>, with an accompanying dashboard for interactive exploration hosted at <https://huggingface.co/spaces/roncmic/crisesStorylinesRAG>. The data is formatted for interoperability and designed to support reuse across research, policy, and operational contexts.

This resource complements existing disaster databases by capturing contextualized relationships between hazards, impacts, and responses that are often absent from conventional event catalogs. Alongside the dataset, we demonstrate a methodology that leverages generative AI to extract and transform information from a large corpus of news articles. This semantic parsing also supports the extraction of place names, which can be linked to geospatial coordinates. In doing so, the resource provides a layer of spatial detail often missing in disaster loss databases, which typically rely on centralized reporting or standardized templates.

Together, these contributions advance a more relational understanding of disaster risk, with applications leveraging HIPs, enhancing early warning systems, and supporting multi-hazard and multi-impact scenario modeling. By transforming diverse textual sources into an well-organized, reusable format, the dataset provides an openly available resource designed for interoperability and reuse in line with the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles²⁷. Beyond research, potential operational advantages include faster data validation and encoding for loss-and-damage tracking, quicker situational awareness and decision support during emergencies, and improved machine-to-machine data exchange. Future extensions could include real-time updates and scenario generation, further strengthening the resource's potential for dynamic monitoring and decision-making in disaster risk management.

Methods

Data Sources for Retrieval. We use two complementary sources to support the information retrieval pipeline: the Emergency Events Database (EM-DAT)¹⁶, which provides a list of disaster events, and the Europe Media Monitor (EMM)^{22,28}, which serves as the primary source of unstructured textual news for generating disaster narratives and knowledge graphs. EM-DAT, maintained by the Centre for Research on the Epidemiology of Disasters (CRED) at University of Louvain, is a globally recognized repository of disaster records, containing over 30,000 entries from 1900 to the present. In this study, EM-DAT is used exclusively as a reference list of events to guide downstream document retrieval. Specifically, we extract the disaster type (e.g., flood, drought), country, and event start date to construct the semantic search queries. This ensures that the retrieval process is anchored into a curated and internationally recognized catalog of disaster occurrences, enabling broad and systematic coverage across hazard types and geographic regions. On the other hand, the EMM, developed by the European Commission's Joint Research Centre (JRC), is a large-scale, multilingual news aggregation and analysis platform. It collects up to 500,000 news reports per day from thousands of verified media sources in over 70 languages, amounting to a historical archive of approximately 1.5 billion articles to date. EMM includes automated tools for language detection, geolocation, event and entity recognition, clustering, deduplication, and semantic topic categorization. Although EM-DAT is used here as a reference list to guide retrieval, the methodology is not limited to this catalog. The same pipeline can be applied to user-defined events-such as a recently reported earthquake

Country	Articles	Unique Sources	% of Articles
United States	2,002,777	2,312	33.19%
United Kingdom	692,280	992	11.47%
India	468,460	127	7.76%
Germany	450,029	368	7.46%
Switzerland	242,700	166	4.02%
France	228,129	167	3.78%
Canada	168,244	148	2.79%
Nigeria	121,274	70	2.01%
Ireland	108,333	159	1.80%
Netherlands	103,584	103	1.72%

Table 1. Top 10 countries by number of English-language articles indexed in the RAG system for 2024. Columns show the total number of articles, the number of unique sources contributing to those articles, and the fraction of all articles represented by each country. This table highlights the concentration of sources in a small number of countries, reflecting coverage and language biases in the dataset.

or flood-by constructing equivalent queries over the EMM corpus. In this way, the present study both enriches EM-DAT records with contextualized information and demonstrates a more generalizable workflow for transforming unstructured disaster reporting into concise, well-organized, and reusable data.

All EMM documents are embedded with the `intfloat/multilingual-e5-large` model (1024-dimensional) on a small GPU cluster (4 × NVIDIA Tesla T4). Inference runs in FP16 to reduce memory footprint and increase throughput without observable loss in retrieval quality. Before embedding, each article is split with a heuristic pipeline built on LangChain’s `RecursiveCharacterTextSplitter` (`chunk_size=1500`, `chunk_overlap=300`, `separators={\n\n, \n., “”}`, `add_start_index=True`). We then apply two post-processing stages to curb noise from very short, low-informativeness fragments: (i) *forward* merging of chunks shorter than 300 characters into their successor whenever the resulting length stays below 1500 chars + 0.5 × 300 chars overflow, and (ii) a *backward* pass that guarantees the final chunk is at least 200 characters by merging it into the penultimate one if necessary. Each chunk retains provenance metadata (original start index, chunk id, length, and number of merges) to support deterministic back-mapping to the source text. To optimize semantic search efficiency, we first apply a keyword-based filter at the article level. Specifically, we restrict the corpus to articles matching 13 EMM categories related to disasters and health crises (e.g., `OHCHR-A1003-EpidemicsPandemicsInfectiousDiseases`, `CommunicableDiseaseOutbreaks`, `NaturalDisasters`, `ManMadeDisasters`, `ChemicalAccidents`,

`NuclearSafety`). Each category is defined by multiple keyword combinations that select relevant articles. This filtering reduces the overall article volume by roughly two orders of magnitude. This yields a total of 11,917 sources, contributing between a few hundred and up to a few million articles per year. The year 2020 is by far the most populated, with about five million articles-most likely reflecting the surge of reporting during the COVID-19 pandemic. The most prolific single source contributes approximately 310,000 articles on its own. Embeddings are stored in an OpenSearch vector index backed by FAISS using an HNSW graph (`m=16`, `ef_construction=100`) with inner-product similarity. We enable symmetric quantization to FP16 (`encoder.name=sq`, `type=fp16`, `clip=true`) to further reduce memory while preserving recall. This setup provides low-latency kNN retrieval over the 1024-D space and serves as the semantic backbone for the downstream RAG stage described above.

The RAG indices span English-language articles drawn from the manually moderated set of sources in EMM, which changed over time as new sources were added to expand domain coverage. We analyzed the geographic distribution of sources indexed in the RAG system for the reference year 2024 (see Table 1). The top ten countries contribute the majority of articles, with the United States, United Kingdom, and India accounting for the largest shares. In total, 185 countries are represented, and the top ten countries alone account for 76% of all indexed articles, while 27 countries account for 90%. The Gini coefficient for source distribution is 0.895, highlighting the strong concentration of articles among a small number of countries. While EMM captures media sources comprehensively, the uneven distribution of vetted sources across countries, combined with our focus on English-language articles, introduces inherent coverage and language biases (see also Technical Validation). These choices optimize retrieval and storyline generation but limit exhaustive global coverage. While they do not alter the conclusions of this study, we recognize these biases as important limitations and opportunities for future research. Expanding the pipeline to additional languages and ensuring more equitable source representation could further enhance the scope, recall, and impact of the dataset.

Generation of Storylines and Graphs. Figure 1 a provides a schematic overview of the workflow used to generate disaster narratives and corresponding knowledge graphs. For each disaster event identified through EM-DAT, we retrieve relevant English-language news articles from the EMM archive to capture both initial impact reporting and early response coverage. This pipeline condenses and organizes high-quality, event-specific information for end-users in DRM and related domains. By combining LLMs with RAG, it extracts semantically relevant content from large-scale news archives and transforms it into interpretable, standardized formats suitable for decision support. EMM’s continuous updates, multilingual scope, and rich metadata make it particularly well suited for retrospective disaster reconstruction and open-domain risk analysis, providing a scalable, real-world foundation for LLM-based disaster data generation.

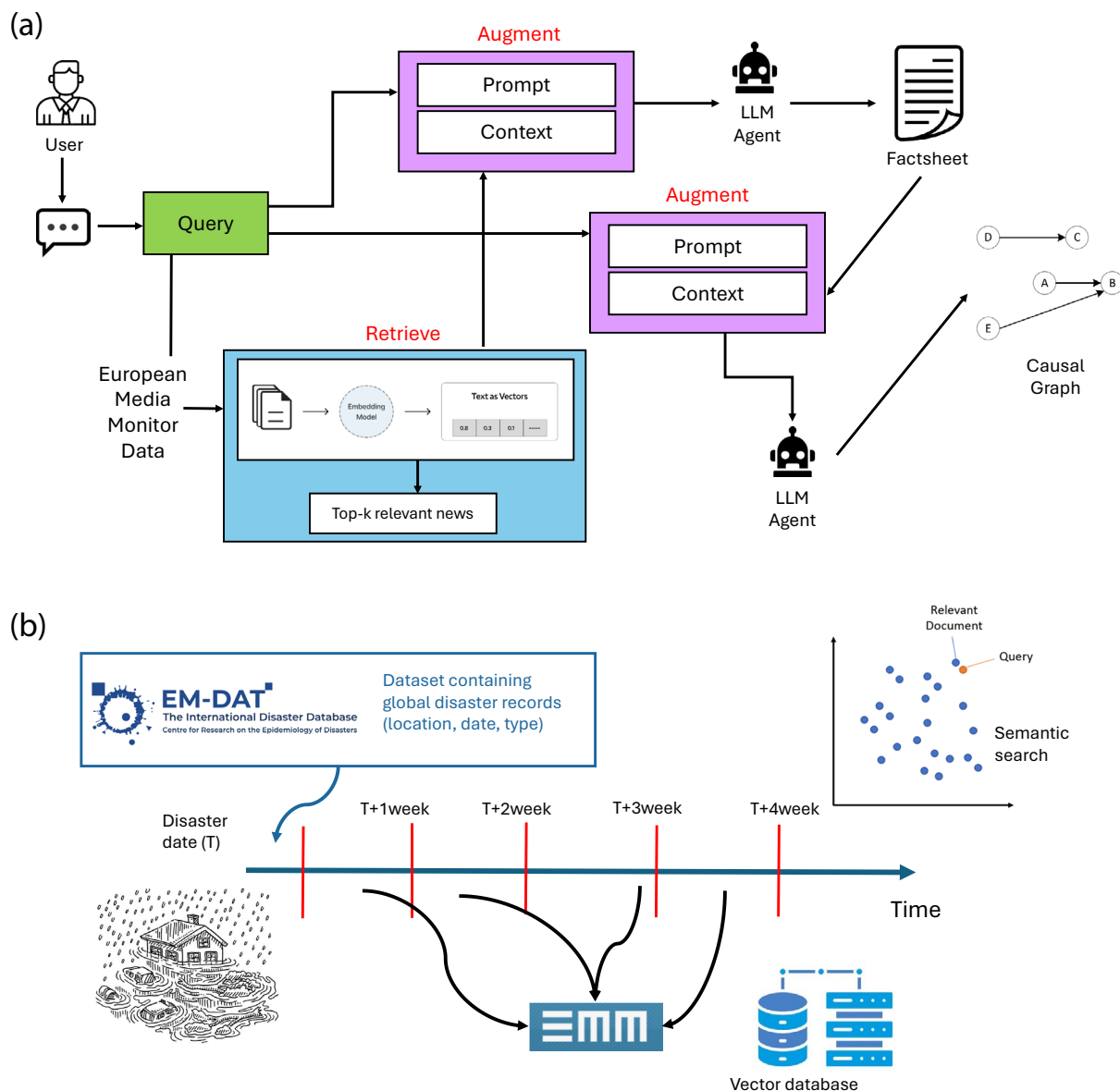


Fig. 1 Pipeline for generating disaster narratives and knowledge graphs from news. **(a)** Schematic of the two-stage generation process. A predefined query is used to retrieve top-k relevant news articles from the EMM. These are passed to a first LLM that generates a structured storyline summarizing the event's key characteristics. A second LLM then converts this storyline into a knowledge graph capturing entities and relationships. **(b)** Querying and filtering process. For each event, semantic search retrieves the top 20 articles for each of the four weeks following the event start. The resulting set is refined by an LLM that filters out unrelated articles to improve relevance and reduce noise.

Starting from the disaster metadata provided by EM-DAT—specifically the event's country, start date, and hazard type—we construct semantic queries to retrieve relevant articles. As explained above, EMM's documents are embedded into separate language-specific indices using chunk-level representations, and in this first experiment we focus on the English-language RAG index, which provides the most uniform and complete coverage across the selected disaster categories, enabling efficient similarity search. For each disaster, we issue four weekly queries in the month following the event's onset, retrieving the top $k = 20$ most relevant articles per query, yielding up to 80 candidate documents per event (Fig. 1b). To reduce noise and improve specificity, we apply an LLM-based filtering step that retains only documents directly referencing the disaster in question. On average, approximately 8.7% of initially retrieved articles pass this filter, significantly enhancing contextual precision for downstream generation. The resulting set of news articles serves as the external knowledge context in a RAG framework, grounding subsequent LLM summaries in semantically relevant evidence.

The filtered document set is then provided as context for the first generation stage, where we prompt the *Meta-Llama-3-70B-Instruct* model²⁹ to generate a disaster storyline or factsheet. This open-weight model, developed by Meta AI, comprises 70 billion parameters and is fine-tuned on a broad range of instructional datasets

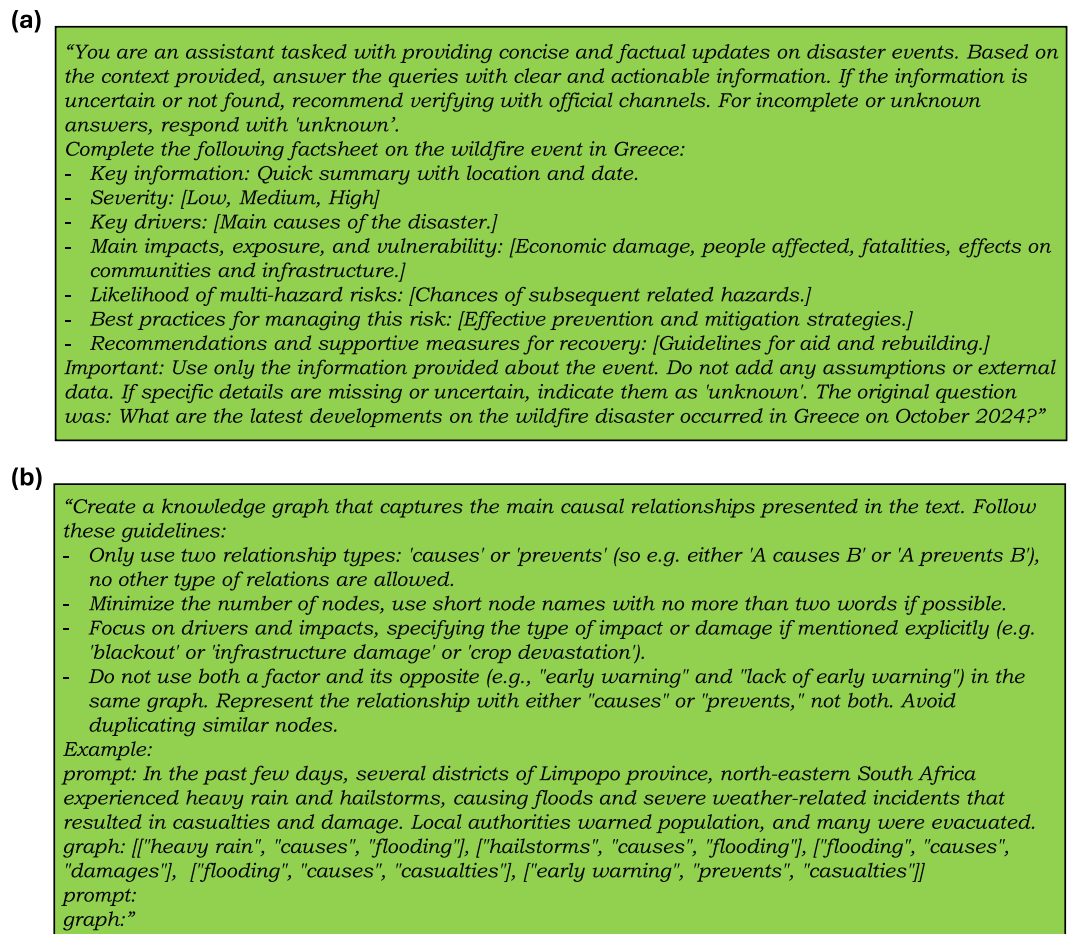


Fig. 2 Prompts used in the AI pipeline for disaster storyline and knowledge graph generation. **(a)** Prompt for generating narrative summaries of disaster events using a set of predefined fields. **(b)** Prompt for extracting KGs using iCL, including an example to guide the generation of subject-predicate-object triples constrained to "causes" and "prevents" relationships.

with extensive human feedback³⁰. It supports a context window of 128,000 tokens and demonstrates strong performance in producing accurate, coherent, and semantically rich summaries across multiple domains²⁹. In our use case, the model is prompted in a zero-shot setting-without additional fine-tuning-to generate summaries that cover key aspects of each disaster event, including core facts, underlying drivers, observed impacts, and reported response actions (see Fig. 2a for the full prompt).

In the second stage, the storyline is passed again to the Llama-3-70B-Instruct model, this time with a prompt tailored to extract the *<subject, predicate, object>* triples that define the main relationships contained in the storyline generated in the first iteration³¹. These triples are used to construct simplified and generalizable KGs centered on disaster dynamics. In this step, the model no longer relies on external retrieval but instead uses the internally generated narrative as contextual input, enabling the extraction of relations already synthesized in the previous stage. To ensure consistency across outputs, the relation types are constrained to two predefined categories: "causes" and "prevents", reflecting reinforcing and mitigating effects, respectively³². This convention is inspired by causal loop diagrams in systems dynamics and DRM modeling^{33,34}. Recent research has shown that LLMs are particularly effective at generating KGs from text via in-context learning (iCL)^{10,35}, which involves providing examples within the prompt to influence the model's output. This approach offers a scalable alternative to traditional graph-based NLP pipelines while preserving semantic meaning across domains³⁶⁻⁴⁴. Examples of extracted triples include *<Heavy rainfall → causes → flash flooding>* or *<Early warning system → prevents → casualties>*. This step also uses zero-shot prompting¹⁰ with iCL, where a formatted example is included in the prompt to guide the structure and output behavior (see Fig. 2b for the full prompt).

Together, this two-stage generation process-organized narratives followed by KGs-leverages the generative capabilities of large-scale LLMs to produce interpretable, scalable outputs for retrospective disaster analysis and decision support. An example output is shown in Fig. 3, including a generated storyline and its associated knowledge graph.

(a)

India-Pakistan heatwave 2015

- **Key information:** A severe heatwave struck Pakistan primarily affecting the Sindh and Punjab provinces in June 2015
- **Severity:** High
- **Key drivers:** Prolonged temperatures above 45°C/113°F for several days, high humidity, power outages which exacerbated the effects of the heat, lack of preparedness and inadequate response by authorities
- **Main impacts, exposure and vulnerability:** Economic damage, people affected, people died and thousands more were affected, fatalities with the majority being elderly or poor. Effects on communities and infrastructure. Hospitals were overwhelmed with heat-related cases. Power outages disrupted daily life and exacerbated the heat. Water scarcity was reported in some areas
- **Likelihood of multi-hazard risk:** Medium to High. The heatwave increased the risk of subsequent hazards such as water scarcity and drought, food insecurity, increased risk of disease transmission e.g. heat-related illnesses waterborne diseases
- **Best management practices:** Establishing early warning systems for heatwaves. Implementing heat action plans including public awareness campaigns and emergency response protocols. Improving access to clean water and electricity. Enhancing healthcare infrastructure and emergency services. Promoting climate-resilient infrastructure and urban planning
- **Recommendations for recovery:** Provide emergency aid including food water and shelter to affected communities. Support the establishment of heat action plans and early warning systems. Invest in climate-resilient infrastructure including green spaces and climate-resilient buildings. Promote public awareness campaigns on heat-related illnesses and prevention measures. Support research and development of climate-resilient technologies and practices. Provide technical assistance and capacity building programs for local authorities and emergency responders

(b)

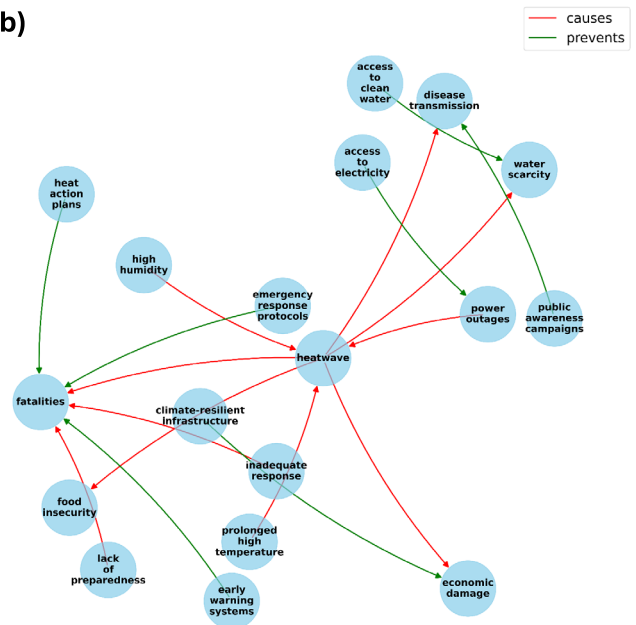


Fig. 3 Example outputs for the 2015 India-Pakistan heatwave. **(a)** Disaster storyline generated by Meta-Llama-3-70B-Instruct, organized into predefined thematic sections. **(b)** Corresponding knowledge graph extracted from the narrative using a second LLM call with iCL, showing causal and preventive relationships constrained to the predicates “causes” and “prevents”.

Data Records

The dataset contains storylines and KGs for 3,158 disaster events occurring between 2014 and 2024 across 175 countries, covering 26 distinct disaster types²¹. Each record corresponds to a unique disaster event, following the EM-DAT structure, identified by the column *DisNo.* (Disaster Number). The dataset is a single CSV file named *DisasterStory.csv*, available at <https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/ETOHA/storylines/DisasterStory.csv>, and also on Zenodo at <https://doi.org/10.5281/zenodo.18598183>. Regular updates are planned on a yearly basis to incorporate new disaster events and improved extraction methods. Each row in the dataset represents a single disaster and includes both standardized metadata and AI-generated analytical content. The storyline for each event is distributed across the following columns: “Key Information”, “Severity”, “Key Drivers”, “Main Impacts, Exposure, and Vulnerability”, “Likelihood of Multi-Hazard Risks”, “Best Practices for Managing This Risk”, and “Recommendations and Supportive Measures for Recovery”. These columns contain the narrative summaries generated using the Meta-Llama-3-70B-Instruct model, synthesizing content primarily from English-language outputs, though the original news sources may be in multiple languages. Each record also includes a “llama graph” column, which encodes a list of <subject, predicate, object> triples extracted from the source articles using LLM-based information extraction. These KG triples represent key relationships (e.g., <strong winds, cause, infrastructure damage>), facilitating reasoning and modeling. The 3,158 records correspond to approximately 50% of all EM-DAT disaster events recorded during the same 2014–2024 period. This partial coverage reflects limitations in the retrieval pipeline, which may fail when relevant news is not present in EMM sources, when semantic search does not surface appropriate articles, or when the LLM-based filtering step excludes non-specific or noisy content. To enhance accessibility, an interactive dashboard is available via Hugging Face Spaces at <https://huggingface.co/spaces/roncmic/crisesStorylinesRAG>, allowing users to explore the dataset by selecting disaster type, country, and event code (*DisNo.*). The platform displays both the storyline and its associated KG in an intuitive and user-friendly interface designed for researchers, practitioners.

Technical Validation

To quantify potential selection biases in our dataset, we compared the events extracted via our RAG pipeline from news to the full EM-DAT record for 2014–2024 (Fig. 4a–d). Coverage is not uniform: sudden-onset disasters such as earthquakes or wildfires are better represented (>80%), while slow-onset events like droughts, heatwaves, or prolonged water shortages are less frequently captured (<20%), likely because our 4-week news-following window and reduced media attention limit detection. Geographically, coverage is highest in Oceania and North America (>80%) and lower in Sub-Saharan Africa, Latin America, and parts of Europe (≤40%), likely reflecting variation in media reporting and English-language availability (see again Table 1). Although the dataset captures roughly half of all EM-DAT events, it covers about 80% of total reported economic losses (see Fig. 4d), reflecting that the most damaging events receive greater media attention. Coverage of fatalities and affected populations is roughly consistent with overall event coverage, and by year exhibits only minor variation.

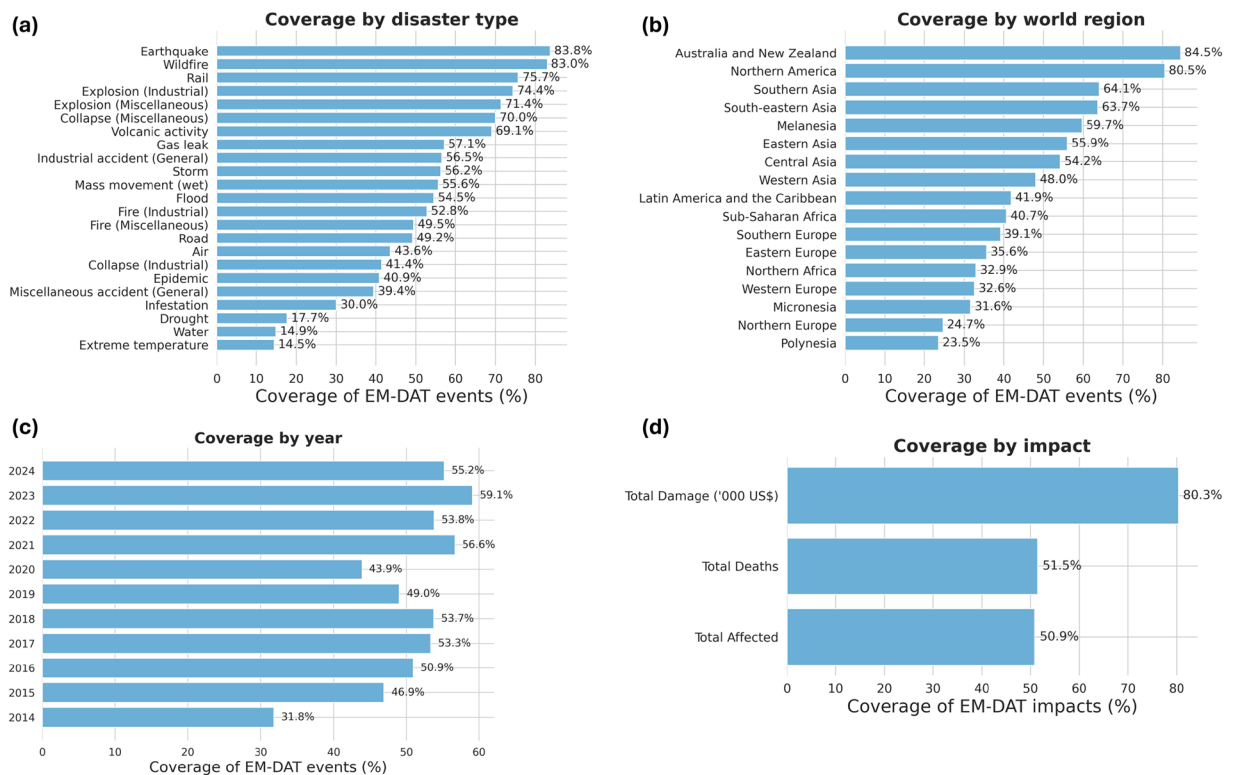


Fig. 4 Coverage of AI-generated narratives and graphs by different dimensions. (a) Coverage by disaster type, showing the fraction of EM-DAT events represented in the story dataset for each type (only disaster types with ≥ 10 events are shown). (b) Coverage by subregion, indicating geographic representation of events. (c) Coverage over time by year, highlighting temporal trends in news reporting. (d) Coverage by impact, showing the fraction of total deaths, affected people, and economic losses captured in the dataset.

While multiple steps in our pipeline—from news retrieval to storyline generation—warrant validation, we focused on the knowledge graphs as a condensed yet information-rich representation of each disaster event. These graphs, built on top of storylines and grounded in retrieved news, offer a fast and interpretable format that captures key relationships between hazards, drivers, impacts, and responses. They thus serve as an effective entry point for expert assessment of the overall quality and factual consistency of the AI-generated outputs. Validating AI-generated content in the DRM domain is inherently challenging. It requires not only technical expertise in hazards and risk modeling but also deep contextual knowledge of sectoral impacts, response phases, and multi-hazard dynamics. As a result, reliable evaluation of such outputs must rely on domain experts with practical and theoretical experience in DRM—resources that are both highly specialized and often difficult to engage. disaster event. Given that, we followed two complementary approaches for validation: one quantitative on a larger sample of triplets, and one qualitative on a small set of full graphs.

For the triplet-based validation, we randomly sampled 1,000 triplets extracted by the LLM from storylines across diverse disaster events, stratified according to the distribution of disaster types in the dataset. Each triplet encodes a causal or factual relationship identified by the LLM from the storyline. Six experts were asked to indicate whether each triplet (e.g., *<heavy rains, cause, flooding>*) is supported by the corresponding text. Given the time-intensive nature of full graph annotation, we focus on precision, as the full set of true relationships per graph is unknown (false negatives cannot be reliably assessed). However, inter-annotator agreement among the six experts was also quantified using Krippendorff's α^{45} and pairwise Cohen's κ^{46} , providing a measure of consistency across evaluations (Fig. 5, Table 2), and indicating a moderate agreement among the experts.

To facilitate the evaluation of KGs, we leveraged the opportunity provided by the “AI for Preparedness” workshop (visit <https://civil-protection-knowledge-network.europa.eu/events/ai-preparedness-building-capacity-ai-powered-drm>), hosted by the European Commission in Brussels from June 16–18, 2025. The event convened around 30 DRM professionals from national civil protection agencies and related response institutions across Member and Participating States of the Union Civil Protection Mechanism. This rare gathering of operational experts created a unique window for direct validation of our outputs. Experts accessed the knowledge graphs through our Hugging Face dashboard (see <https://huggingface.co/spaces/roncmic/crisesStorylinesRAG>) and viewed both the storyline and its corresponding graph to ensure context-based evaluation. Participants reviewed a sample of 34 KGs, selected to reflect high-impact, thematically diverse events and aligned with each expert's domain of expertise. Using a predefined rubric, they rated each graph according to the following criteria: “Fully Correct” (all key nodes and causal/preventive links are accurate), “Mostly Correct” (minor issues but generally sound), “Partially Incorrect” (some key elements wrong or missing), “Incorrect” (major inaccuracies), or “Difficult to Judge” (insufficient context or ambiguity). Experts were given approximately 40 minutes to complete the task, relying on their own

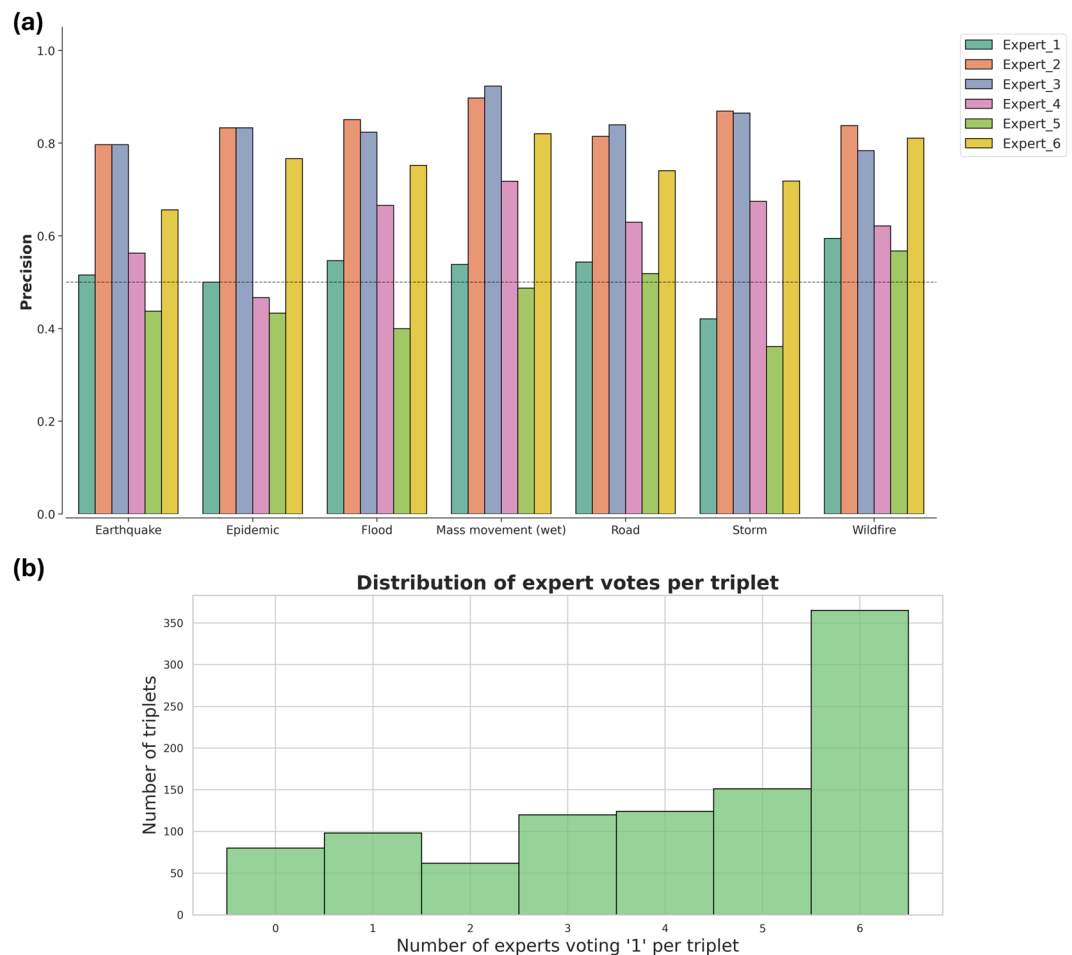


Fig. 5 (a) Precision of LLM-generated triplets per expert, disaggregated by disaster type (only types with >20 triplets). (b) Distribution of expert agreement per triplet (0–6 experts).

Metric	Value
Krippendorff's α (all experts)	0.425
Average pairwise Cohen's κ	0.451
Minimum pairwise Cohen's κ	0.207
Maximum pairwise Cohen's κ	0.665
Majority vote Precision	0.640

Table 2. Summary of inter-annotator agreement and overall precision of expert evaluations on LLM-generated triplets. Krippendorff's α and pairwise Cohen's κ quantify agreement across all six experts. Majority vote Precision indicates the fraction of triplets labeled as “True” by the majority of experts. The distribution of agreement counts (0–6 votes per triplet) is visualized in Fig. 5.

knowledge and any external sources they deemed necessary. Evaluations were submitted directly via the dashboard and automatically logged. While the sample represents a small portion of the full dataset, the focused input of this specialized group provides meaningful first insights into the quality of the AI-generated graphs and helps guide future improvements.

To summarize the validation outcomes, we aggregated expert evaluations across disaster types. Since multiple experts could assess the same event, each event received a set of ratings, allowing us to compute the relative distribution of evaluation categories per disaster type. Table 3 reports the percentage of assessments falling into each rating class—Fully Correct, Mostly Correct, Partially Incorrect, Incorrect, and Difficult to Judge—along with the total number of evaluated graphs per hazard type. These aggregated results provide an overview of perceived accuracy across hazards and help identify areas where the AI-generated outputs performed more reliably or raised interpretation challenges. This analysis not only highlights strengths in our current approach but also pinpoints specific areas needing refinement to enhance the reliability and utility of the AI-generated content in the DRM domain.

Disaster Type	Difficult to Judge	Fully Correct	Incorrect	Mostly Correct	Partially Incorrect	N
Drought	0.0%	0.0%	5.0%	87.4%	7.5%	7
Earthquake	0.0%	0.0%	0.0%	0.0%	100.0%	2
Epidemic	0.0%	70.0%	0.0%	15.2%	14.9%	2
Explosion (Industrial)	0.0%	0.0%	0.0%	24.4%	75.6%	2
Flood	2.3%	36.2%	2.1%	37.3%	22.2%	14
Storm	0.0%	0.0%	0.0%	94.3%	5.7%	3
Volcanic activity	0.0%	0.0%	0.0%	28.3%	71.7%	1
Water	0.0%	0.0%	0.0%	100.0%	0.0%	1
Wildfire	0.0%	0.0%	32.7%	0.0%	67.3%	2

Table 3. Distribution of expert assessments by disaster type, following EM-DAT categories¹⁶. Values represent the percentage of all individual expert ratings in each category. The final column indicates the number of unique disaster graphs evaluated for each type.

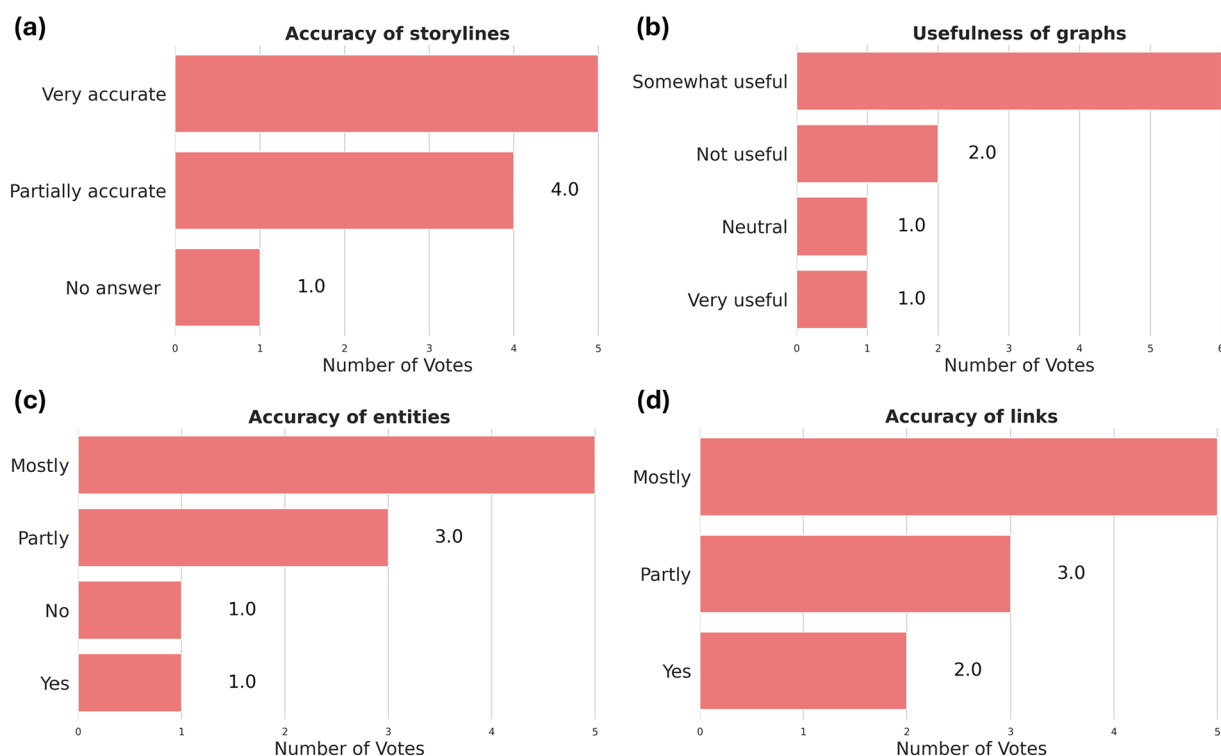


Fig. 6 Results of the survey assessing the AI-generated narratives and graphs. **(a)** Accuracy of Storylines (options: “Very accurate,” “Partially accurate,” and “Inaccurate.”) **(b)** Usefulness of the Graphs (options: “Very useful,” “Somewhat useful,” “Neutral,” and “Not useful.”) **(c)** Accuracy of Entities (options: “Yes, all accurately identified and represented,” “Mostly, but a few inaccuracies are present,” “Partly, some entities are correctly identified while others are not,” “No, many entities are inaccurately identified or missing,” and “Not sure / I don’t have enough information to assess.”) **(d)** Accuracy of Links/Relationships (options: “Yes, all accurately identified and represented,” “Mostly, but a few inaccuracies are present,” “Partly, some links are correctly identified while others are not,” “No, many links are inaccurately identified or missing,” and “Not sure / I don’t have enough information to assess.”).

In addition to direct graph evaluation, we designed a short EU survey (available at <https://ec.europa.eu/eusurvey/runner/AI4DRM>) to gather broader, overall feedback on the quality and usefulness of the AI-generated disaster storylines. Ten participants completed the survey, representing diverse roles including Civil Protection Managers, DRM Practitioners, Researchers, and Policy Makers. We report results for four targeted questions, each assessed after participants interacted with the Hugging Face dashboard. These included: (1) “How accurate did you find the AI-generated storylines on disaster events?” (2) “How useful did you find the AI-generated knowledge graphs for understanding disaster narratives?” (3) “Are the entities in the graph accurately identified and represented?” and (4) “Are the relationships between entities accurately extracted and represented?”. Figure 6 summarizes responses in four histograms. Main findings include that half of respondents rated the storylines as “Very accurate,” more than half found the graphs “Somewhat useful,” and approximately

half considered both the entities and links to be “Mostly accurate”. These results provide encouraging evidence of perceived quality and relevance from an operational DRM perspective, even if based on a relatively small group of experts. Since all data are openly available and we provide an interactive Hugging Face dashboard to facilitate exploration, our approach is intended as a fully open-source model. This setup allows end users to engage directly with the storylines and graphs, potentially enabling larger-scale validation and refinement through broader community interaction.

Usage Notes

This dataset can support a range of applications in DRM, including retrospective analysis, multi-hazard risk assessment, scenario generation, and simulation of cascading effects. By transforming large volumes of unstructured news content into structured storylines and KGs, it bridges textual information and formal risk models. The data can complement existing hazard inventories such as EM-DAT, offering narrative and relational detail often absent from conventional disaster databases. Relationships linking drivers, impacts, and response measures can inform vulnerability assessments, early warning strategies, and integration into system dynamics or Bayesian network models. Practitioners may use the interactive dashboard to explore past events, gaining context-specific insights into secondary effects, compounding drivers, and effective interventions. Researchers in climate adaptation, humanitarian coordination, or environmental security may benefit from the narratives, which capture contextual nuance such as infrastructure, governance, or pre-existing vulnerabilities shaping disaster outcomes. The dataset also provides a unique resource for AI research. To our knowledge, it is the first large-scale collection of disaster-related KGs paired with multi-sentence narratives, enabling benchmarking in text-to-graph and graph-to-text generation, causal reasoning, complex factual summarization, and multimodal representation learning. It can serve as a testbed for evaluating alignment, coherence, and factual accuracy in generative models grounded in graph-based representations of factual information.

Users should be aware of several limitations. All storylines and KGs are derived from publicly available news sources, which vary in factual reliability, regional coverage, and thematic emphasis. This introduces potential geographic and content biases, including underrepresentation of events in low-income regions or slow-onset hazards. The driving factors identified reflect reported narratives and may not correspond to underlying systemic drivers. The zero-shot LLM-based generation process, while scalable, can produce incomplete or imprecise representations, especially for complex or under-reported events. KGs are simplified by design, capturing only two relation types, and are not intended as comprehensive causal models. The dataset is seeded on EM-DAT and thus inherits its known incompleteness; some disasters are missing¹⁶. However, the pipeline is fully generalizable, allowing users to generate storylines and graphs from alternative disaster inventories, news sources, or user-defined sets of events. For high-stakes applications, outputs should be triangulated with additional sources or reviewed by domain experts.

This release is fully open: code, models, prompts, and processed outputs are publicly available, ensuring methodological transparency and enabling adaptation to new sources or contexts. Beyond providing openly accessible data, this work introduces a novel methodology that combines large-scale news retrieval, embedding-based retrieval augmentation, and LLM-driven extraction of disaster knowledge graphs and storylines. While the current dataset represents a first demonstration rather than a definitive disaster knowledge repository, it establishes the feasibility of generating structured, interpretable representations from unstructured news. Future work may include more extensive validation, integration of additional data modalities (e.g., time series, official statistics), multilingual coverage, probabilistic reasoning modules (e.g., Bayesian networks), and regular updates to capture emerging events. Overall, the dataset offers a scalable, interpretable, and reproducible foundation for multi-hazard awareness, disaster risk management research, and AI-driven analysis.

Data availability

The dataset containing both storylines and KGs is available in CSV format within the Joint Research Centre Data Catalogue at <https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/ETOHA/storylines/DisasterStory.csv>²¹. To maximize access and visibility, we also make all data and code available on Zenodo in a single repository, which can be downloaded at <https://doi.org/10.5281/zenodo.18598183>.

Code availability

All Python code used for data processing, RAG pipeline, and knowledge graph extraction is available at <https://github.com/jrcf7/crisesStorylinesRAG>. The code for the Gradio application can be found at <https://huggingface.co/spaces/roncmic/crisesStorylinesRAG>. All data and code are also made available on Zenodo in a single repository, which can be downloaded at <https://doi.org/10.5281/zenodo.18598183>.

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Competing interests

The authors declare no competing interests.

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